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The quant2qual pipeline

From numeric distributions to qualitative case studies, one stage at a time

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Reference: quant2qual pipeline guide

Abstract

quant2qual translates behavioral metrics (time, activity and friction) into case-study prose. This guide walks through the four stages of the workflow on the synthetic cohort shipped with the package: configure the domain, classify every case against an archetype matrix, narrate the cases and the cohort, and compile the report. Every figure and table is produced by the function named beside it, so each page can be reproduced in one line.

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1 The shape of the workflow

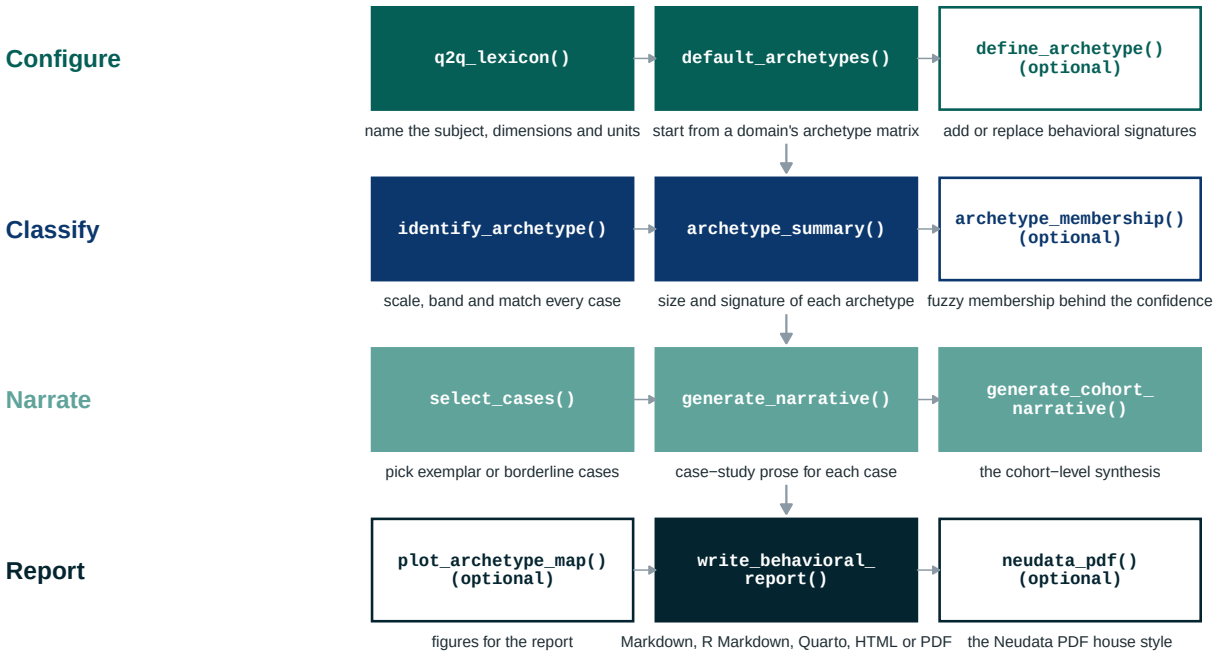


Figure 1: The four stages: plot_q2q_pipeline()

Read it top to bottom. The first row is configuration: **the domain is described once**, as a lexicon (what the subject, the dimensions and their units are called) and an archetype matrix (which behavioral signatures exist). Every later call reads those two objects, which is why the same script can narrate a usability study, a course or a survey without editing a single function. The outlined boxes are optional.

Table 1: The pipeline as a table: q2q_pipeline()

Stage	Function	Required	What it does
Configure	q2q_lexicon()	required	name the subject, dimensions and units
Configure	default_archetypes()	required	start from a domain's archetype matrix
Configure	define_archetype()	optional	add or replace behavioral signatures
Classify	identify_archetype()	required	scale, band and match every case
Classify	archetype_summary()	required	size and signature of each archetype
Classify	archetype_membership()	optional	fuzzy membership behind the confidence
Narrate	select_cases()	required	pick exemplar or borderline cases
Narrate	generate_narrative()	required	case-study prose for each case
Narrate	generate_cohort_narrative()	required	the cohort-level synthesis
Report	plot_archetype_map()	optional	figures for the report
Report	write_behavioral_report()	required	Markdown, R Markdown, Quarto, HTML or PDF
Report	neudata_pdf()	optional	the Neudata PDF house style

2 Configure: describe the domain once

Every dataset is read on three behavioral dimensions. The lexicon names them in the language of the domain; the engine never sees the names, only the numbers.

Table 2: Three domain packs, one geometry: q2q_lexicon()

Domain	Subject	Velocity	Intensity	Friction
ux	participant	session duration	interaction volume	error events
learning	learner	time on task	learning activity	failed attempts
survey	respondent	completion time	open-text elaboration	data-quality flags

The **archetype matrix** lists the signatures and the bands each dimension must fall in. Friction-led signatures are checked first, then the four time/activity quadrants, and whatever matches nothing goes to the cohort-typical fallback.

Table 3: The UX archetype matrix: default_archetypes("ux")

Archetype	Velocity	Intensity	Friction	Theme
Frustrated Fighter	mid|high	high	high	obstructed persistence
Frustrated Novice	low|mid	low|mid	high	early exit under friction
Methodical Explorer	high	low|mid	low|mid	deliberate depth
Power Navigator	low|mid	high	low|mid	fluent expertise
Passive Disengaged	low	low	low|mid	shallow contact
Baseline User	–	–	–	cohort-typical behavior

New signatures are added with `define_archetype()` and combined with `archetype_matrix()`; each carries its own persona, headline, interpretation and recommendations, so the narrative engine needs no further configuration.

3 Classify: scale, band, match

`identify_archetype()` standardises every dimension against the cohort, cuts it into low, mid and high bands, and matches each case to an archetype. Because the bands are relative, "high friction" automatically rescales to whatever dataset is plugged in – unless fixed raw-unit `cutpoints` are given.

Table 4: Band edges in raw units at +/- 0.5 SD: identify_archetype()

Dimension	Median	Low band	High band
Velocity (time)	23	< 17.0	> 35.5
Intensity (actions)	60	< 43.7	> 95.9
Friction (errors)	2	< 1.2	> 5.4



Figure 2: Where every case sits: plot_archetype_map()

Every case also receives a **fuzzy membership** in every archetype – a smooth version of "inside the allowed bands", 0.5 exactly on a band edge. The membership of the assigned archetype is the case's **confidence**; the next-best archetype is its **runner-up**. On this synthetic cohort the planted signatures are known, so the classifier can be checked directly: it recovers 84% of them.

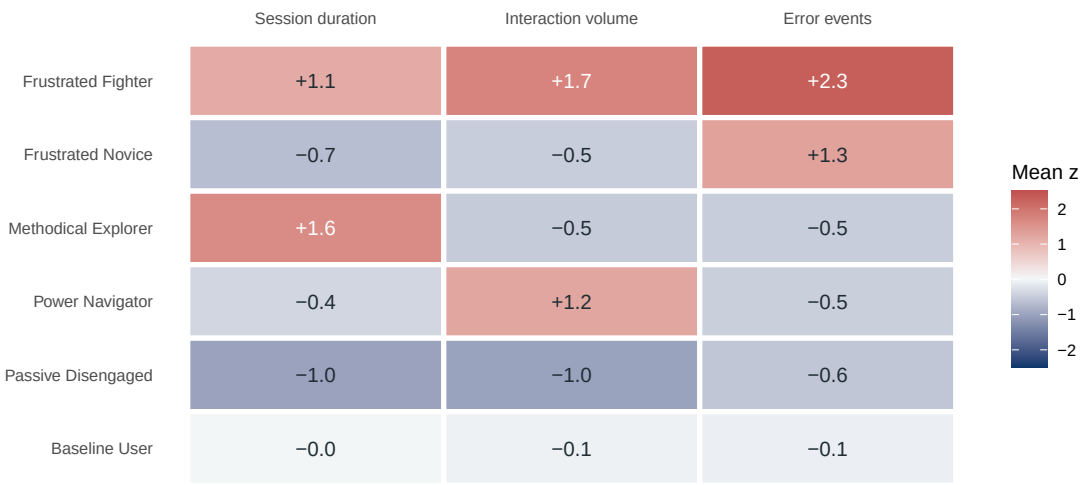


Figure 3: What makes each archetype distinctive: plot_archetype_signatures()

4 Narrate: cases and cohort

`generate_narrative()` writes a case study from the case's own statistics. Sentences are assembled from templates whose slots are filled with percentiles, cohort ratios and standardised scores, ordered by salience and stitched with connectors that follow the direction of the data. Phrasing varies between cases but is fixed for each one: the same data always produce the same text.

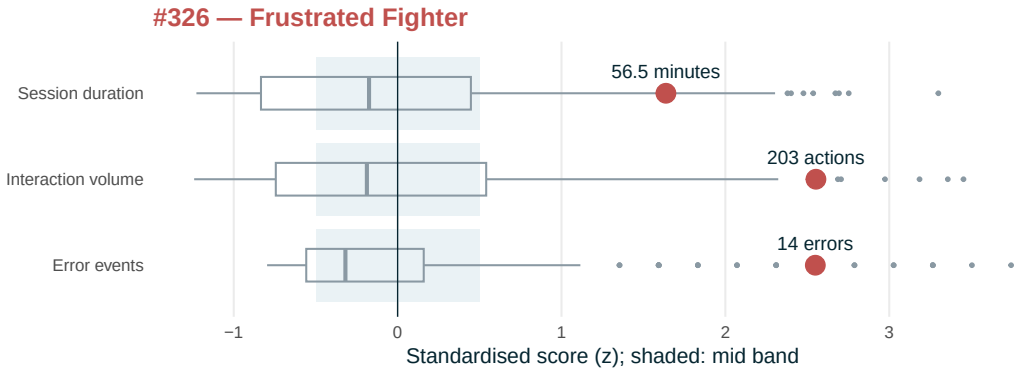


Figure 4: One case against the cohort: plot_case_profile()

i Case Study ID: #326 – generate_narrative()

Frustrated Fighter — Obstructed persistence · confidence high (1.00)

Presenting pattern. The participant exhibited a state of high interaction friction. Taken as a whole, the record is most consistent with the Frustrated Fighter archetype — an engaged but deeply stuck user struggling against system barriers, exhibiting high friction and repetitive loops.

Behavioral evidence. Interaction volume sat within the upper 4th percentile (203 actions against a cohort median of 60 actions; z = +2.55), indicating a dense, rapid stream of interface actions. Compounding this, error events were far above the cohort norm at 14 errors (about 7 times the cohort median; z = +2.55), consistent with a recurring pattern of failed or rejected actions. In parallel, session duration was markedly above the cohort norm at 56.5 minutes (about 2.5 times the cohort median; z = +1.64), consistent with an extended investment of time in the product. Normalising for exposure, the record shows an interaction cadence of 3.6 actions per minute (about 40% above the cohort median) and an error rate of 6.9 errors per 100 actions (roughly twice the cohort median). On interaction volume and error events, the participant lies beyond the Tukey fence of the cohort distribution and is a statistical outlier; values of this kind merit verification against the raw logs before interpretation.

Interpretation. Taken together, the profile describes effortful persistence under obstruction rather than disengagement: the participant kept working at the task while the interface repeatedly pushed back. Rapid, consecutive actions in the presence of frequent errors are characteristic of interface confusion and rising cognitive load, and the length of the session suggests that the goal mattered enough to keep trying. Left unaddressed, this pattern is a leading indicator of churn among otherwise motivated users.

Considerations

- Replay sessions from this group around the first error to locate the step where the loop begins.
- Audit error messages on that path for actionability: each should state what went wrong and what to do next.

- Add an in-context recovery aid (inline validation, undo, or a guided fallback) that triggers after repeated errors.

Confidence and caveats. Classification confidence is high (membership 1.00): the record is a clear example of the Frustrated Fighter signature. All comparisons are relative to the present cohort of 240 participants (z scaling); the same raw values could read differently against another reference group.

The same case in the other two registers:

Brief. The participant exhibited a state of high interaction friction. Taken as a whole, the record is most consistent with the Frustrated Fighter archetype — an engaged but deeply stuck user struggling against system barriers, exhibiting high friction and repetitive loops. Interaction volume sat within the upper 4th percentile (203 actions against a cohort median of 60 actions), indicating a dense, rapid stream of interface actions. Compounding this, error events were far above the cohort norm at 14 errors (about 7 times the cohort median), consistent with a recurring pattern of failed or rejected actions. Confidence: high (1.00).

UX, recommended actions.

- Replay sessions from this group around the first error to locate the step where the loop begins.
- Audit error messages on that path for actionability: each should state what went wrong and what to do next.
- Add an in-context recovery aid (inline validation, undo, or a guided fallback) that triggers after repeated errors.

`generate_cohort_narrative()` writes the synthesis that opens a report: the distribution of archetypes, the friction-led share, how concentrated friction is, how the dimensions relate, and how cleanly the cohort was classified.

Cohort synthesis – generate_cohort_narrative()

The cohort comprised 240 participants. Their behavior resolved into 6 recognisable signatures. The most common was the Baseline User profile (30% of the cohort), characterised by cohort-typical behavior. It was followed by Passive Disengaged (21%), Power Navigator (17%), Methodical Explorer (15%), Frustrated Fighter (9%) and Frustrated Novice (8%).

Taken together, 17% of participants fall into friction-led profiles (Frustrated Fighter and Frustrated Novice): the group in which design or support intervention is most likely to change outcomes. The distribution of error events was uneven: the top 10% of participants accounted for 42% of all errors, while 21% recorded none at all.

Across the cohort, greater session duration was moderately associated with greater interaction volume (Spearman's rho = 0.41); greater session duration was weakly associated with more error events (Spearman's rho = 0.26); greater interaction volume was moderately associated with more error events (Spearman's rho = 0.30).

Median classification confidence was 0.96. 13 participants (5%) were classified with low confidence and sit between signatures, most often between Baseline User and Methodical Explorer. 37 participants (15%) lie beyond a Tukey fence on at least one dimension.

`select_cases()` chooses which cases to narrate: the clearest example of each archetype (`"representative"`), the most pronounced (`"extreme"`), or the cases sitting between signatures (`"borderline"`), which are the best audit of a matrix.

5 Report: one call, five formats

Table 5: Report targets: write_behavioral_report()

Format	Output	Needs
markdown	pandoc Markdown text or file	nothing
rmd	R Markdown source using neudata_pdf	nothing
qmd	Quarto source (HTML and PDF)	nothing
html	self-contained HTML, Neudata palette	pandoc
pdf	PDF in the Neudata LaTeX house style	pandoc and LaTeX

```
write_behavioral_report(profile, file = "study.pdf", format = "pdf",
                        title = "Checkout usability study",
                        client = "Example Ltd", confidentiality = "Confidential")
```

A report contains key findings, the cohort synthesis with its figures and archetype portraits, a case study per selected case with its fingerprint chart, and a methods section written from the classification settings – the scaling, the band edges in raw units and the full archetype matrix. This guide itself is typeset with `neudata_pdf()`, the same output format.

Produced with quant2qual 0.1.0 on 15 October 2026. Source: `inst/pipeline/quant2qual-pipeline.Rmd`. Rebuild with `rmarkdown::render(system.file("pipeline", "quant2qual-pipeline.Rmd", package = "quant2qual"))`.



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